

# Skills Summary

## Basic Matrix Operations

Use this reference while completing your worksheet or when studying.

**Size first, always.** A matrix with  $m$  rows and  $n$  columns is  $m \times n$ . Addition and subtraction need *identical* sizes; a product  $AB$  needs the columns of  $A$  to equal the rows of  $B$ , and then  $AB$  is (rows of  $A$ )  $\times$  (columns of  $B$ ).

**Entry by entry:**  $(A \pm B)_{ij} = a_{ij} \pm b_{ij}$  and  $(kA)_{ij} = k a_{ij}$ .

**Row meets column:**  $(AB)_{ij}$  is row  $i$  of  $A$  paired term by term with column  $j$  of  $B$ , then added. Order matters —  $AB$  and  $BA$  are usually different, and one may not even exist.

**Determinant** ( $2 \times 2$ ):  $ad - bc$  for  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ . **Equal matrices** have equal corresponding entries.

### 1. Adding, subtracting and scaling

Check the sizes match, then work position by position. A scalar reaches *every* entry — the most common slip is multiplying only the first row. Nothing here mixes entries from different positions.

**Example:** Find  $A + B$  for  $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$ ,  $B = \begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$ .

**Step 1.** Both matrices are  $2 \times 2$ , so the sum exists and every entry of the answer comes from the matching pair of entries — no row-column pairing is involved.

**Step 2.**  $2 + 5 = 7$ ,  $-1 + 2 = 1$ ,  $4 + (-3) = 1$ ,  $3 + 1 = 4$ .  $A + B = \begin{bmatrix} 7 & 1 \\ 1 & 4 \end{bmatrix}$

**Try it:** Find  $3A$  for  $A = \begin{bmatrix} 2 & -5 \\ 1 & 4 \end{bmatrix}$ .

$\begin{bmatrix} 6 & -15 \\ 3 & 12 \end{bmatrix}$  :check:

### 2. Multiplying: row meets column

Slide row  $i$  of the left matrix down column  $j$  of the right one, multiplying the pairs you meet and adding the results. That single number is the entry in row  $i$ , column  $j$ . If the row and the column have different lengths, the product does not exist.

**Example:** Find  $AC$  for  $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$  and the column  $C = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ .

**Step 1.**  $A$  is  $2 \times 2$  and  $C$  is  $2 \times 1$ : the inner dimensions agree, so the answer is a  $2 \times 1$  column — one number per row of  $A$ .

**Step 2.** Row 1:  $(1)(4) + (2)(2) = 8$ . Row 2:  $(3)(4) + (1)(2) = 14$ .  $AC = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$

**Try it:** Find  $AC$  for  $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$  and  $C = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ .

$\begin{bmatrix} 5 \\ 11 \end{bmatrix}$  :check:

### 3. Determinants and matrix equations

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The  $2 \times 2$  determinant is main diagonal minus anti-diagonal:  $ad - bc$ . Subtract, never add. For a matrix equation, equal matrices means equal corresponding entries, so one matrix statement becomes one ordinary equation per position.

**Example:** Find the determinant of  $\begin{bmatrix} 6 & 4 \\ 3 & 5 \end{bmatrix}$ .

**Step 1.** A single number is wanted from a  $2 \times 2$  matrix, and the determinant formula is the only operation that does that: main diagonal product minus the other one.

**Step 2.** Main diagonal  $(6)(5) = 30$ ; anti-diagonal  $(4)(3) = 12$ ; subtract. The determinant is **18**, and being nonzero it says the matrix is invertible.

**Try it:** Find  $x$  if  $\begin{bmatrix} x-3 & 4 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 2 & 7 \end{bmatrix}$ .

check:  $11 = x$

(!) **Watch out:** matrix multiplication is not commutative — compute  $AB$  and  $BA$  separately, and expect different answers or a product that does not exist at all. When scaling, reach every entry. When taking a determinant, subtract the second product rather than adding it.